

SUMMARY

I am a 3rd-year PhD candidate in MAMBO team at Toulouse University (expected graduation: Dec. 2026), advised by [Prof. Pierre Weiss](#) and [Prof. Edouard Pauwels](#). My research focuses on computational imaging with deep learning and generative models. I also actively contribute to open-source software.

WORK EXPERIENCE

**Research Engineer – [Agenium Space](#)** Dec 2022 - Sept 2023

Correcting micro-vibrations in satellite imaging with pushbroom cameras. Modeling the imaging process, implementing deep learning-based methods, and evaluating them on real data from Sentinel-2B.

**Research Intern – [Toulouse Mathematics Institute](#)** Jun 2022 - Sept 2022

Product-convolution neural networks: applications to image deblurring with space-varying blurs, with [Pierre Weiss](#) and [Paul Escande](#).

EDUCATION

2023 - present PhD (Applied Mathematics, Inverse problems in Imaging) at **Toulouse University**  
2018 - 2023 Master of Engineering in Applied Mathematics at **INSA Toulouse** (Major in AI / Data Science)

OPEN-SOURCE PROJECTS

**Blind deblurring in Fluorescence Microscopy** Napari plugin released soon!

Designing deep learning-based methods to jointly estimate the sharp image and the unknown blur kernel from a single blurred fluorescence microscopy image. We address diffraction-limited blurs based on Zernike polynomials, spatially varying blurs and 2D/3D settings. Efficient implementation in PyTorch and ongoing validation on various microscopy settings (RIM, TIRF, confocal). Joint work with [Pierre Weiss](#), [Florian Sarron](#), [Paul Escande](#) and opticians [Thomas Mangeat](#), Sylvain Cantaloube. Presented at [MIFOBIO-25](#).

**DeepInverse – Core contributor and maintainer** [Link to Github](#)

An open-source PyTorch-based library (part of the [PyTorch Ecosystem](#)) for solving imaging inverse problems with state-of-the-art deep learning based methods. `deepinv` accelerates research across imaging domains, enhances research reproducibility via a common modular framework of problems and algorithms. Responsibilities: core library API design/contribution, benchmarks, CI/CD, packaging, docs, and community support.

SKILLS

|                    |                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python: PyTorch (advanced), triton/cuda kernel (intermediate), JAX. C++ (intermediate)<br>Profiling/optimization (Nsight, torch-profiler)                                                 |
| <b>Development</b> | Git/Github, packaging (PyPI), virtual environments ( <code>conda</code> , <code>uv</code> , <code>docker</code> ).<br>SOTA implementation and evaluation, model training and fine-tuning. |
| <b>Research</b>    | Image restoration, computational imaging, deep learning, deblurring/denoising                                                                                                             |
| <b>Language</b>    | English (professional), French (fluent), Vietnamese (native)                                                                                                                              |

TEACHING

- Teaching Assistant (Vacataire) – **INSA Toulouse** 2023 - present  
Courses: Signal Processing, Stochastic Optimization

## DISTINCTIONS

---

- French National Prize for Open Science Library for DeepInverse Nov 2024
- Doctoral Research Fellowship, French Ministry of Higher Education and Research 2023 - 2026
- IA Pau Winner 2023

## PUBLICATIONS

---

Tachella, Julián et al. (2025). *DeepInverse: A Python package for solving imaging inverse problems with deep learning*. arXiv: [2505.20160](#) [[eess.IV](#)].

Nguyen, Minh Hai and Pierre Weiss (2024). “Comparing Plug-and-Play and Unrolled networks”. In: *Pre-print*.

Nguyen, Minh Hai et al. (2024). “DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras”. In: *IEEE Transactions on Geoscience and Remote Sensing* 62, pp. 1–9.

Nguyen, Minh Hai et al. (2024). “How Diffusion Prior Landscapes Shape the Posterior in Blind Deconvolution”. In: *Pre-print*.